DATA SCIENCE FOR BUSINESS · CHAPTER 2

Data, Information and Business Questions

The DIKW model · Data sources · Research design · Measurement scales · Turning vague problems into measurable analytical questions



Data Science Fundamentals for Business School Students

WISDOM *definition: The ability to use your* ***knowledge*** *and* ***experience*** *to make* ***good decisions and judgments***

Where Chapter 2 Fits

Chapter 1 recap

Data science turns raw data into insight that supports decisions.

Data can be **structured** (tables), **semi-structured** (JSON, logs) or **unstructured** (text, images, audio).

The lifecycle always begins with a business question.

Chapter 2 focus

Before any analysis starts, we must agree on:

- what counts as data
- where it comes from
- how it is measured
- which question it should answer

Why it matters

Poor questions lead to poor data; poor data leads to poor decisions.

Most failed analytics projects fail at the **start** — the question was unclear, the data did not fit, or success was never defined.

Big message Data science is not only about algorithms. The hardest business skill is turning a concern such as **“sales are falling”** into a testable question such as **“Which customer segments, channels and product categories explain the sales decline over the last 12 months?”**

BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO

Learning Objectives

1. **Explain** the DIKW model: how Data becomes Information, Knowledge and Wisdom.
2. **Distinguish** internal vs external data, and primary vs secondary data.
3. **Compare** observational and experimental data in business contexts.
4. **Classify** variables and measurement scales: nominal, ordinal, interval, ratio.
5. **Convert** vague management problems into measurable analytical questions.
6. **Identify** the unit of analysis, target variable, data source and success metric.



Assessment alignment You will be assessed on **explaining, classifying, designing questions and justifying data choices** — not only on memorising definitions. Every activity in this deck practises one of these skills.

LECTURE STRUCTURE

Today’s Roadmap

PART 1

DIKW model

How raw facts become business **decisions**.

PART 2

Data sources

Internal vs **external**; **primary** vs **secondary**.

PART 3

Research designs

Observational vs **experimental** data.

PART 4

Variables

Variable types and **measurement** scales.

PART 5

Analytical questions

From vague **problems** to **measurable questions**.

PART 6

Practice & assessment

Knowledge checks, challenges and a **short test**.

01

PART 1

From Data to Business Wisdom

The DIKW model — and why raw facts are not yet decisions

IN THIS PART

- Warm-up: four statements
- The four DIKW levels
- A retail DIKW ladder
- Common mistakes
- Knowledge check and activity

Warm-up: What Is the Difference?

Data

"Customer 1042 bought Product A at 14:35 for £29.99."

What makes it data? *A single **recorded** fact. No comparison, no meaning yet.*

Information

"Product A sold 340 units this week, 18% more than last week."

What makes it information? *Facts are **aggregated** and **compared** over time.*

Knowledge

"Sales rise when Product A is placed near the checkout."

What makes it knowledge? *A **pattern** that explains why something happens.*

Wisdom

"Move Product A near the checkout before weekends."

***What makes it wisdom?** A **decision** about what to do next.*

Discuss in pairs Which of the four is most useful to a manager? Which is easiest to collect? Why are they not the same thing?

The DIKW Model

Business interpretation

The DIKW model explains why data does not automatically create value. **A spreadsheet is not a decision.** Data needs *context*, *interpretation* and *action* before it helps a business.

An analyst moves up the model by asking **three** questions:

- **What does this mean?** (data → information)
- **Why did it happen?** (information → knowledge)
- **What should we do next?** (knowledge → wisdom)

Each step adds context and value — and requires more judgement.



Level 1: Data

Definition

Data are **raw facts or observations** recorded before they are organised into a meaningful form.

Data can be *numbers, text, clicks, timestamps, images, survey answers, transaction records, sensor readings or social-media posts*.

Typical characteristics: unprocessed, high in volume, low in context, and often generated automatically by business systems.

Business examples

- 10,000 website clicks
- A receipt line: £29.99, Product A
- Customer age: 24
- Store ID: 17
- A call-centre audio recording
- A rating: 3 out of 5
- A delivery timestamp

Key point Data may be **accurate but still meaningless** without context. “24” could be an age, a store number or a delivery time — only context tells us which.

Level 2: Information

Data example

“£29.99” — a number with no context.

Information example

“Average basket value increased from £24.60 to £29.99 after the weekend promotion.”

What context was added?

What?
Average basket value

When?
After the weekend promotion

Who / where?
All customers in the store

Compared with?
£24.60 before the promotion

Management use Information is data organised with context. It helps managers **describe what is happening** — reports and dashboards live here — but it does not always explain **why** it is happening.

Level 3: Knowledge

Knowledge means

Understanding **patterns** in information: *relationships, causes, segments, probabilities, risk factors, drivers and constraints.*

It answers the question **“why?”** or **“under what conditions?”**

Example

Information: sales rose after the promotion.

Knowledge: the increase was strongest among returning customers using mobile checkout.

The second statement identifies **who** drove the change and **how**.

Why it matters

Knowledge helps a manager judge whether a pattern is **repeatable, valuable and worth acting on**.

Without it, managers may copy a result that happened by chance.

Caution Knowledge is not certainty. It is an **evidence-based interpretation** that must be tested and updated as new data arrives.

Level 4: Wisdom

Wisdom in business

Wisdom is **choosing an appropriate action** using data, information and knowledge, while weighing:

- costs and expected return
- risks and uncertainty
- ethics and fairness
- timing
- organisational constraints

Example

If mobile promotions increase returning-customer conversion **without reducing profit margin**, a wise action might be:

- run targeted weekend mobile campaigns
- monitor margin, churn and customer satisfaction
- stop or adjust if any metric deteriorates

Key distinction A model may **predict**; managers must still **decide responsibly** — and remain accountable for the decision.

Figure: The DIKW Ladder in a Retail Example



Common DIKW Mistakes

Mistake 1: “More data is always better”

Irrelevant data adds noise, cost, storage and confusion.

Better practice: Collect data that answers the business question.

Mistake 2: “A dashboard is knowledge”

A dashboard displays information. It becomes knowledge only when interpreted against a question, context and decision.

Better practice: Pair every dashboard with the question it answers.

Mistake 3: “Correlation is wisdom”

A pattern is not automatically a good action. Business constraints, ethics and uncertainty still matter.

Better practice: Test causes and consider risks before acting.

Mistake 4: “Data removes judgement”

Data improves judgement; it does not replace managerial responsibility.

Better practice: Combine evidence with experience and ethics.

Coming next A knowledge check — be ready to classify statements into DIKW levels.

Knowledge Check 1 — DIKW

Classify each statement as Data, Information, Knowledge or Wisdom — then justify your answer.

Statement	Level	Why?
“Customer ID 772 clicked the email link at 08:41.”		
“Email click-through rate was 7.2% this week.”		
“Students opening emails before 9am are more likely to attend tutorials.”		
“Schedule reminder emails for 8am and monitor attendance.”		

Knowledge Check 1 — DIKW

Classify each statement as Data, Information, Knowledge or Wisdom — then justify your answer.

Statement	Level	Why?
“Customer ID 772 clicked the email link at 08:41.”	**Data**	A single raw event with no comparison or meaning.
“Email click-through rate was 7.2% this week.”	**Information**	Events aggregated into a metric for a period.
“Students opening emails before 9am are more likely to attend tutorials.”	**Knowledge**	A relationship (pattern) between behaviour and outcome.
“Schedule reminder emails for 8am and monitor attendance.”	**Wisdom**	A chosen action, with monitoring built in.



Pair discussion Why is the third statement **not yet wisdom**? What would need to be added to turn it into a decision?

PART 1 · GROUP ACTIVITY · 5–8 MINUTES

Activity 1 — Build a DIKW Ladder

Scenario A

A coffee shop wants to increase **repeat visits**.

Scenario B

An online retailer sees more **abandoned baskets**.

Scenario C

A university wants to improve **first-year attendance**.

Your task: complete all four levels for one scenario

DATA

A raw fact the business records

INFORMATION

An organised summary or comparison

KNOWLEDGE

A pattern or likely cause

WISDOM

An action and how you will monitor it

02

PART 2

Data Sources

Where business data comes from — and how source choices shape decisions



IN THIS PART

- What a data source is
- Internal and external sources
- Evaluating source quality
- Primary and secondary data
- Activity: choosing sources

What Is a Data Source?

Definition

A data source is the **origin of the data** used for analysis.

It can be a *system, database, survey, sensor, transaction log, interview, public dataset, web platform or third-party provider.*

Knowing the source tells you **how, why and by whom** the data was created — and therefore what it can and cannot tell you.

Why the source matters

Accuracy — Is it correct?

Bias — Who is missing?

Cost — What does it cost to get?

Access — Can we legally use it?

Privacy — Does it contain personal data?

Timeliness — Is it recent enough?

Fit — Does it match our question?

Managerial question Can we trust this data **enough to make this decision** from it?

Internal Data Sources

Sales / POS

Transactions, basket value, returns, discounts, product codes.

CRM

Customer profiles, contact history, complaints, loyalty activity.

Operations

Inventory, delivery times, staffing, service wait times.

Digital

Web clicks, app events, email opens, search terms.

Finance

Invoices, revenue, costs, cash flow, budgets.

HR

Attendance, training, recruitment, turnover, engagement.

Key idea Internal data is usually **cheaper and more specific**, but it can reflect internal blind spots and historical bias — it only shows the customers and processes you already have.

External Data Sources

Market and competitor data

Industry reports, competitor prices, market-size estimates, trade publications.

Public and open data

Government statistics, census data, economic indicators, transport and weather data.

Social and web data

Reviews, social posts, search trends; web scraping only where legal and ethical.

Third-party data

Data brokers, credit bureaux, commercial datasets, syndicated surveys.

Caution External data can be valuable, but always check **source credibility, licensing, legality and sampling bias** before relying on it.

Internal vs External Data

Dimension	Internal data	External data
Main strength		
Typical cost		
Control		
Bias risk		
Business example		

Management takeaway Strong business analysis usually **combines** internal and external sources: internal data shows what happened to us; external data shows what happened around us.

Internal vs External Data

Dimension	Internal data	External data
Main strength	Specific to the organisation	Adds market or environmental context
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Business example	CRM churn history	Competitor pricing or census demographics



Management takeaway Strong business analysis usually **combines** internal and external sources: internal data shows what happened to us; external data shows what happened around us.

Evaluating a Data Source: A Quality Checklist

Relevance

Does the data measure what our question needs?

Accuracy

Were values recorded correctly and consistently?

Completeness

Are important records or fields missing?

Timeliness

Is it recent enough for this decision?

Representativeness

Does it cover the population we care about?

Legality and ethics

Do we have permission and a lawful basis to use it?

Rule of thumb If a source fails **relevance or legality**, do not use it — no amount of cleaning can fix those problems.

Primary Data

Definition

Primary data is **collected directly for the current purpose** or research question.

It is designed by the organisation or researcher to meet a **specific decision need**, so the variables, sample and timing are chosen by you.

Examples

- A customer survey designed this semester
- Interviews with store managers
- A focus group about a new product
- An A/B test on a website
- Direct observation of queue times

Advantages

High relevance and control; up to date; you know exactly how it was collected.

Trade-offs

Requires time, budget, careful design and ethical consent; samples are often small.

Secondary Data

Definition

Secondary data **already exists** and was collected by someone else, or for another purpose.

The analyst **reuses** it to answer a new question — so it is important to check how the original collector defined and measured each variable.

Examples

- Government labour statistics
- Published financial statements
- Old CRM exports
- Industry reports
- Previous survey results
- Public health datasets

Advantages

Faster and cheaper; often large, broad coverage; useful for benchmarks and trends.

Trade-offs

May not match your exact question, population or definitions; may be outdated.

Primary vs Secondary Data

Question	Primary data	Secondary data
Who collected it?		
Main advantage		
Main risk		
Example		
Best use		

Best practice Use **secondary data to frame** the problem, then collect **primary data** when the decision needs more precise evidence.

Primary vs Secondary Data

Question	Primary data	Secondary data
Who collected it?	You, your organisation or your research team	Someone else, or another system
Main advantage		
Main risk		
Example		
Best use		



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Example	Survey current customers about delivery preferences	Census income and local footfall data
Best use		



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Main risk	Cost, poor survey design, small sample	Mismatch, outdated, unknown bias
Example	Survey current customers about delivery preferences	Census income and local footfall data
Best use	When the question is new or specific	When context or a benchmark is needed



Best practice Use **secondary data to frame** the problem, then collect **primary data** when the decision needs more precise evidence.

PART 2 · PAIR ACTIVITY · 8 MINUTES

Activity 2 — Source Choice

Problem 1

A bank wants to predict which customers may **close their accounts** in the next 3 months.

Problem 2

A retailer wants to decide whether to **open a new store** in a different city.

Problem 3

A university wants to understand why first-year students **miss seminars**.

For each problem, identify…	Your answer
One internal source — and what it tells you	
One external source — and what it adds	
Do you need primary data? Why or why not?	

03

PART 3

Observational vs Experimental Data

How the way data is collected affects what we can conclude



IN THIS PART

- Observational data
- Experimental data and A/B tests
- When experiments are difficult
- Bias and confounding
- Knowledge check

Observational Data

Definition

Observational data is collected **without deliberately controlling** who receives a treatment or intervention.

We simply **observe what happens** in the real world — customers, employees and markets behave as they normally would.

Why it is common: business systems already record it automatically and cheaply.

Examples

- Customer purchase history
- Employee turnover records
- Website clickstream logs
- Store footfall by day
- Complaints by region
- Historical sales and weather

Main limitation Observational data can show **association**, but causal claims are difficult because other factors may explain the pattern.

Experimental Data

Definition

Experimental data is collected when an organisation **deliberately applies an intervention** and compares outcomes between groups.

Examples

- A/B test on a website
- Two prices tested in comparable stores
- Training programme assigned to one team
- Email subject lines randomly assigned

Four building blocks of an experiment

Treatment
The change being tested

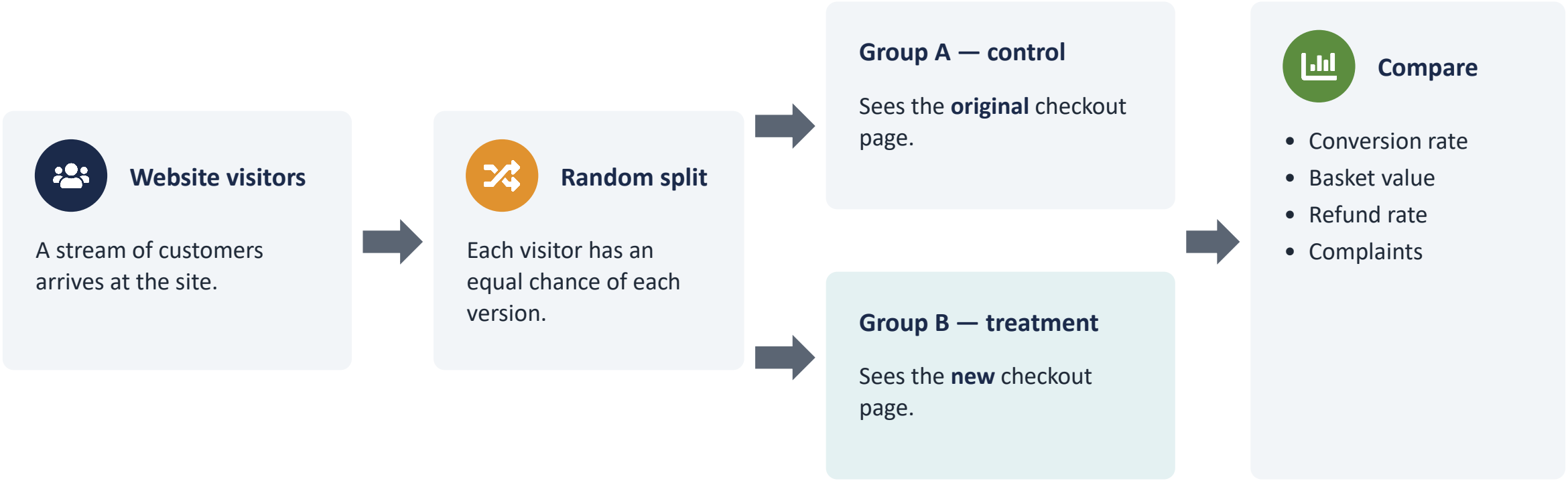
Control group
Receives no change, for comparison

Random assignment
Chance decides who gets what

Outcome variable
The result we measure

Main strength Well-designed experiments provide the **strongest evidence about cause and effect**.

Figure: A/B Testing Logic



Key rule **Random assignment** makes the groups comparable, so differences in outcomes are more likely to be caused by the new page.

When Experiments Are Difficult

Ethical limits

You cannot deliberately expose customers, patients or employees to harmful conditions just to collect data.

Cost and time

Some experiments are too expensive, too slow or too disruptive to normal operations.

Operational constraints

Managers may refuse random assignment if all customers must receive the same service.

External validity

A controlled test may not represent real market conditions, other regions or other seasons.

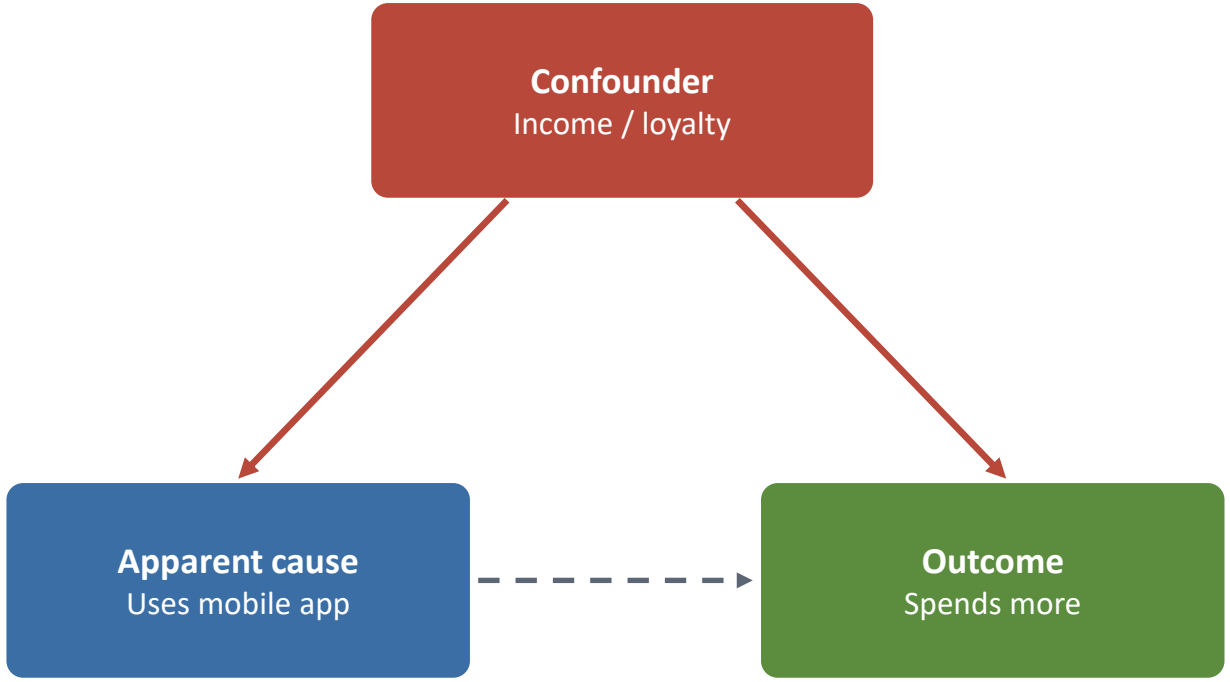
Business implication Know what your design allows you to claim: **description**, **prediction** or **causation**. Only strong designs support causal claims.

Bias and Confounding

Confounding

A **confounder** is a third factor related to both the apparent cause and the outcome. It can make a relationship look causal when it is not.

Example: customers who use the mobile app spend more. But higher-income, loyal customers are more likely to install the app. **Loyalty or income** may explain both.



Observed link — is it causal?

Key phrase Association is **not** the same as causation.

Knowledge Check 2 — Research Design

Scenario	Design	Why?
Comparing historical sales in rainy vs sunny weeks	**Observational**	Nobody controls the weather; we only observe.
Randomly showing half of website visitors a new homepage	**Experimental**	The business assigns the treatment at random.
Surveying customers about service satisfaction	**Observational**	Opinions are recorded; no intervention is applied.
Assigning half of stores a new shelf layout and comparing sales	**Experimental**	A deliberate change with a comparison group.

Reflection Which scenarios support **stronger causal claims**, and what could still go wrong in them (e.g., stores that are not comparable)?

04

PART 4

Variables and Measurement Scales

What exactly are we measuring?

IN THIS PART

- What a variable is
- Categorical vs numerical
- Nominal, ordinal, interval, ratio
- Why measurement choices matter
- Activity: classify the variables

What Is a Variable?

Definition

A variable is a **characteristic that can take different values** across people, products, transactions, stores, departments, days or other units of analysis.

Examples

Age · gender · revenue · satisfaction score · product category · number of visits · delivery time · churn status

Variable, value and observation in a dataset

Customer ID	Region	Visits	Satisfaction
C001	North	12	4
C002	South	3	**2**

Variable = a column (e.g., Satisfaction)

Value = one cell (highlighted: 2)

Observation = one row (customer C002)

Why it matters: the type of variable determines which statistics, charts and models are appropriate.

Categorical vs Numerical Variables

Categorical variables

Values are **labels or groups**.

- **Nominal:** no order — region, payment method, department
- **Ordinal:** ordered — risk level, satisfaction band

Examples: product category, customer segment, churn yes/no.

Analyse with: counts, percentages and bar charts.

Numerical variables

Values are **numbers representing quantities**.

- **Discrete:** counted — number of visits, items in basket
- **Continuous:** measured — revenue, delivery time, age

Examples: profit margin, spend per customer.

Analyse with: mean, median, spread, histograms and line charts.

Knowledge check Is customer satisfaction from 1 to 5 categorical or numerical? It is **ordinal** — but often averaged as if numerical. Discuss why this is tricky.

Measurement Scale 1 — Nominal

✗ Order | ✗ Equal distances | ✗ True zero

Nominal scale

Categories with **names but no natural order**.

You can count how often each category appears, but “higher” or “lower” **does not make sense**.

Numbers used as codes (North = 1, South = 2) are still nominal.

Business examples

- Payment method: card / cash / transfer
- Region: north / south / east / west
- Product category: food / clothing / electronics
- Customer type: new / returning

Appropriate summaries Counts, percentages, **mode**, bar charts. Avoid means and line charts.

Measurement Scale 2 — Ordinal

✓ Order | ✗ Equal distances | ✗ True zero

Ordinal scale

Categories have a **meaningful order**, but the **distance between** levels is **not guaranteed to be equal**.

We know “high” is above “medium”, but not by how much.

Most survey rating questions (Likert scales) produce ordinal data.

Business examples

- Satisfaction: very dissatisfied → very satisfied
- Risk: low / medium / high
- Service rating: 1 to 5 stars
- Education level
- Priority: urgent / normal / low

Caution The gap between 1 and 2 stars may not equal the gap between 4 and 5 stars. Prefer **median**, ranks and **% by level**.

Measurement Scale 3 — Interval

✓ Order | ✓ Equal distances | ✗ True zero

Interval scale

A numerical scale where **differences are meaningful**, but **zero does not mean "none"**, so ratios are not meaningful.

20°C is 10 degrees warmer than 10°C — but it is **not "twice as hot"**.

The zero point is arbitrary: it was chosen by convention, not by nature.

Business examples

- Temperature in °C or °F
- Calendar years
- Some index scores (e.g., confidence indices)
- Standardised test scores

Business use Interval measures appear in indices and some survey scales. Means and differences work; **ratios do not**.

Measurement Scale 4 — Ratio

✓ Order | ✓ Equal distances | ✓ True zero

Ratio scale

A numerical scale with **equal intervals and a true zero**.

Both differences and ratios are meaningful: “twice as much”, “50% growth” and “half the time” **all make sense**.

Most business KPIs — sales, costs, time, counts — are ratio variables.

Business examples

- Revenue and profit
- Age
- Number of customers
- Delivery time
- Units sold
- Website visits

Interpretation £200 revenue is **twice** £100 revenue; zero sales means no sales. Mean, SD, ratios and growth rates are all valid.

Comparison Table — Measurement Scales

Scale	Order?	Equal distance?	True zero?	Example	Useful summaries
Nominal	No	No	No	Payment method	Counts, percentages, mode
Ordinal	**Yes**	Not guaranteed	No	Satisfaction 1–5	Median, rank, % by level
Interval	**Yes**	**Yes**	No	Temperature °C	Mean, SD, differences
Ratio	**Yes**	**Yes**	**Yes**	Revenue, age, units	Mean, SD, ratios, growth

Rule of thumb Before analysing any variable, ask: **what kind of variable is this?** Each scale adds one property to the one above it — and unlocks more valid summaries.

Why Measurement Choices Matter

Weak measurement

A company measures **“customer loyalty”** only by number of purchases.

This ignores customers who love the brand but buy rarely because products are **expensive or seasonal** — and it rewards frequent buyers who may be price-hunters with no loyalty at all.

Better measurement

Combine several indicators:

- Repeat purchase and time since last purchase
- Referral behaviour
- Satisfaction and complaint history
- Customer lifetime value

Managerial lesson **What you measure shapes what you manage.** A narrow metric can push staff to optimise the wrong behaviour.

PART 4 · INDIVIDUAL, THEN GROUPS · 8 MINUTES

Activity 3 — Classify the Variables

Variable	Type (categorical / numerical)	Measurement scale	Suitable chart
Customer gender			
Satisfaction from 1 to 5			
Monthly revenue			
Region			
Delivery time in minutes			
Risk category (low / medium / high)			

Challenge For each variable, name **one suitable chart** and **one unsuitable chart**, and explain why the unsuitable one would mislead.

PART 4 · INDIVIDUAL, THEN GROUPS · 8 MINUTES

Activity 3 — Classify the Variables

Variable	Type (categorical / numerical)	Measurement scale	Suitable chart
Customer gender	Categorical	Nominal	Bar chart
Satisfaction from 1 to 5			
Monthly revenue			
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Delivery time in minutes			
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PART 4 · INDIVIDUAL, THEN GROUPS · 8 MINUTES

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Customer gender	Categorical	Nominal	Bar chart
Satisfaction from 1 to 5	Categorical	Ordinal	Ordered bar chart
Monthly revenue			
Region			
Delivery time in minutes			
Risk category (low / medium / high)			



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PART 4 · INDIVIDUAL, THEN GROUPS · 8 MINUTES

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Variable	Type (categorical / numerical)	Measurement scale	Suitable chart
Customer gender	Categorical	Nominal	Bar chart
Satisfaction from 1 to 5	Categorical	Ordinal	Ordered bar chart
Monthly revenue	Numerical, continuous	Ratio	Histogram; line chart over time
Region			
Delivery time in minutes			
Risk category (low / medium / high)			



Challenge For each variable, name **one suitable chart** and **one unsuitable chart**, and explain why the unsuitable one would mislead.

PART 4 · INDIVIDUAL, THEN GROUPS · 8 MINUTES

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Region	Categorical	Nominal	Bar chart
Delivery time in minutes			
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PART 4 · INDIVIDUAL, THEN GROUPS · 8 MINUTES

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Monthly revenue	Numerical, continuous	Ratio	Histogram; line chart over time
Region	Categorical	Nominal	Bar chart
Delivery time in minutes	Numerical, continuous	Ratio	Histogram or box plot
Risk category (low / medium / high)			



Challenge For each variable, name **one suitable chart** and **one unsuitable chart**, and explain why the unsuitable one would mislead.

PART 4 · INDIVIDUAL, THEN GROUPS · 8 MINUTES

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Customer gender	Categorical	Nominal	Bar chart
Satisfaction from 1 to 5	Categorical	Ordinal	Ordered bar chart
Monthly revenue	Numerical, continuous	Ratio	Histogram; line chart over time
Region	Categorical	Nominal	Bar chart
Delivery time in minutes	Numerical, continuous	Ratio	Histogram or box plot
Risk category (low / medium / high)	Categorical	Ordinal	Ordered bar chart



Challenge For each variable, name **one suitable chart** and **one unsuitable chart**, and explain why the unsuitable one would mislead.

05

PART 5

From Vague Problems to Analytical Questions

The core business skill of data science

IN THIS PART

- Why vague questions fail
- What makes a good question
- Four types of business questions
- Unit, target, granularity, time window
- Three worked examples

The Problem: Managers Often Start Vague

"Sales are bad."

"Customers are unhappy."

"Marketing is not working."

Why vague questions fail — they leave out six things:

Who?
Which customers or units

What?
Which metric

When?
Which period

Where?
Which market or channel

How much?
What size of change

Compared to what?
Which baseline

Without these, it is impossible to choose the right **data source, variable, method or success metric**.

Goal of this section Turn a **management concern** into a **measurable analytical question**.

What Makes a Good Analytical Question?

Specific

Defines the population, business area and timeframe.

Measurable

Uses variables and metrics that can be observed or collected.

Actionable

Connects to a decision the organisation can actually make.

Comparable

Includes a baseline, benchmark or comparison group.

Feasible

Can be answered with the available time, skills and data.

Ethical

Can be investigated responsibly, fairly and legally.

Checklist If the question fails any of these tests, **refine it before collecting data** — it is far cheaper to fix a question than a finished analysis.

A Template for Writing Analytical Questions

For [population] **during [timeframe]** **what explains / predicts [target metric]** **compared with [baseline]** **so that we can [decision]?**

Worked example

For **first-time online customers** during **the last two quarters**, what predicts **a second purchase within 60 days**, compared with **the same period last year**, so that we can **target welcome offers more efficiently?**

Why it works

Every element points to a specific variable, data source or comparison — so the analyst knows exactly what data to collect.

Common error

Leaving out the decision. A question without a “so that…” often produces interesting findings that nobody uses.

Four Types of Business Questions

Type	Question asked	Business example	Typical output
Descriptive	What happened?	What were monthly sales by region?	Report / dashboard
Diagnostic	Why did it happen?	Why did sales fall in Region A?	Driver analysis
Predictive	What is likely to happen?	Which customers are likely to churn?	Forecast / probability
Prescriptive	What should we do?	Which offer should we send to each segment?	Recommended action

Analytics maturity: value and difficulty increase →



Most organisations mature from descriptive towards prescriptive analytics over time.

Question Refinement Funnel



How to use the funnel

1. Write down the concern exactly as the manager said it.

2. Narrow the scope: who, what and when.

3. Choose a metric that captures the concern.

4. Check which data sources contain that metric.

5. Write the final question — it should name the population, timeframe and target.

Define the Unit of Analysis

Definition

The unit of analysis is the **“thing” each row or case represents**: a customer, transaction, store, product, employee, patient, student, day or campaign.

Ask: **“What am I making a decision about?”** — that is usually your unit.

Examples

- Customer-level churn model
- Transaction-level fraud model
- Store-level sales comparison
- Student-level attendance analysis
- Product-level inventory forecast

Why it matters If the unit is unclear, variables get **double-counted or mixed across levels**, and conclusions become confused.

Define the Target Variable and Success Metric



Target variable

The **outcome you want to explain or predict**. It sits inside the dataset.

Examples: churn yes/no, monthly revenue, exam pass/fail, number of complaints, delivery delay.

It answers: “What are we trying to explain or predict?”



Success metric

How you judge whether the analysis or intervention was useful to the business.

Examples: reduce churn by 5%, improve forecast accuracy, increase profit, reduce waiting time.

It answers: “Did the project help the business?”



Business warning A model can be **statistically accurate but commercially unhelpful** if the success metric is wrong.

Granularity and Time Window

Granularity

The **level of detail** in the data: transaction, daily, weekly, monthly, customer, store or product level.

Time window

The **period covered**: last week, last quarter, last 12 months, before/after a campaign, or a test period.

Example: “Sales are falling” can look different at each level

Day | Week | Month | Product | Store | Segment

Guideline Choose a window and granularity that **match the decision**: weekly staffing needs daily data; annual strategy needs multi-year monthly data.

Worked Example 1 — Sales Decline

Vague problem

"Sales are falling."

Better question

"Which product categories and customer segments explain the fall in monthly revenue over the last 12 months?"

Output

A ranked driver analysis by category, segment and channel.

Element	Decision
Unit of analysis	Monthly sales by product category × segment × channel
Target variable	Monthly revenue
Data sources	POS, CRM, promotion calendar, web analytics
Comparison	This year vs last year; before vs after campaign
Success metric	Identify actionable drivers explaining at least 80% of the decline

Worked Example 2 — Customer Churn

Vague problem “Too many customers are leaving.”

Design choice	Churn example
Analytical question	
Unit of analysis	
Target variable	
Predictors	
Data sources	
Business action	
Success metric	

Worked Example 2 — Customer Churn

Vague problem “Too many customers are leaving.”

Design choice	Churn example
Analytical question	Which active customers are most likely to stop buying in the next 90 days?
Unit of analysis	Customer
Target variable	Churn within 90 days: yes / no
Predictors	Purchase history, complaints, tenure, discounts, channel, satisfaction
Data sources	CRM, transaction records, support tickets, survey
Business action	Retention offer, service call, loyalty incentive
Success metric	Lower churn rate and positive ROI of the retention campaign

Worked Example 3 — Student Retention

Vague problem “Some first-year students are struggling.”

Design choice	Student retention example
Analytical question	
Unit of analysis	
Target variable	
Predictors	
Data sources	
Ethical caution	
Success metric	

Worked Example 3 — Student Retention

Vague problem “Some first-year students are struggling.”

Design choice	Student retention example
Analytical question	Which first-year students are at risk of non-progression by week 6, and which support factors are associated with improvement?
Unit of analysis	Student
Target variable	Progression / non-progression, or an attendance risk score
Predictors	Attendance, VLE activity, assessment submission, support meetings, commute distance
Data sources	Attendance system, VLE logs, student records, support services
Ethical caution	Avoid labelling students unfairly; use for support, not punishment
Success metric	Improved attendance, progression and student satisfaction

Knowledge Check 3 — Improve the Question

Vague problem A

"Our advertising is not effective."

Vague problem B

"Our employees are leaving."

Task: for one problem, write a question that includes…

- Population
- Timeframe
- Target variable
- Data sources
- Success metric

Model answer style "Which marketing channels generated **profitable new customers** among **18–30-year-olds** in the **last quarter**, after controlling for campaign spend?" — it names a population, timeframe, target and comparison, and points directly to the data needed.

06

PART 6

Practical, Activities and Assessment

Applying the chapter to realistic business scenarios

IN THIS PART

- Question-design canvas
- Data source map
- Three graded challenges
- End-of-lecture test
- Key takeaways

PART 6 · WORKSHEET

Practical Canvas — Question Design

Canvas item	Your answer
1. Vague business problem	
2. Decision to be made	
3. Unit of analysis	
4. Target variable / outcome	
5. Key predictors / explanatory variables	
6. Data sources needed	
7. Time window and comparison group	
8. Success metric	
9. Ethical or data-quality risk	



How to use it

Complete the canvas **before collecting data**.

Work top to bottom: each answer should follow logically from the one above.

If you cannot fill in an item, the question needs more refinement.

Most often forgotten: items 2 and 8.

PART 6 · WORKSHEET

Practical Canvas — Question Design

Canvas item	Your answer
1. Vague business problem	Too many orders arrive late, causing customer complaints.
2. Decision to be made	
3. Unit of analysis	
4. Target variable / outcome	
5. Key predictors / explanatory variables	
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PART 6 · WORKSHEET

Practical Canvas — Question Design

Canvas item	Your answer
1. Vague business problem	Too many orders arrive late, causing customer complaints.
2. Decision to be made	Decide whether to change courier or improve warehouse dispatch procedures.
3. Unit of analysis	
4. Target variable / outcome	
5. Key predictors / explanatory variables	
6. Data sources needed	
7. Time window and comparison group	
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Practical Canvas — Question Design

Canvas item	Your answer
1. Vague business problem	Too many orders arrive late, causing customer complaints.
2. Decision to be made	Decide whether to change courier or improve warehouse dispatch procedures.
3. Unit of analysis	One customer order.
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5. Key predictors / explanatory variables	Courier, destination region, dispatch delay, order size, shipping method and day of the week.
6. Data sources needed	Order records, warehouse dispatch logs, courier tracking and promised delivery dates.
7. Time window and comparison group	Analyse the past six months; compare couriers within similar regions and shipping methods.
8. Success metric	
9. Ethical or data-quality risk	



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7. Time window and comparison group	Analyse the past six months; compare couriers within similar regions and shipping methods.
8. Success metric	Reduce the late-delivery rate by at least 20% relative to baseline during an eight-week pilot, without increasing cost per order by more than 5%.
9. Ethical or data-quality risk	



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7. Time window and comparison group	Analyse the past six months; compare couriers within similar regions and shipping methods.
8. Success metric	Reduce the late-delivery rate by at least 20% relative to baseline during an eight-week pilot, without increasing cost per order by more than 5%.
9. Ethical or data-quality risk	Protect customer addresses; check missing timestamps and inconsistent definitions of “delivered.” Courier comparisons may be biased by different routes.



How to use it

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PART 6 · GROUP ACTIVITY · 15 MINUTES

Practical Activity — Data Source Map

1

Choose a problem

Churn, low sales, absenteeism, late delivery, complaints or fraud.

2

List internal sources

CRM, sales system, website, HR, finance, operations or support logs.

3

Add external / primary

Surveys, interviews, government data, competitor data, weather or market reports.

4

Map sources to the question

Link each source to the part of the question it answers.

Deliverable A **one-page source map** showing which data answers which part of the question, with each source labelled internal/external and primary/secondary.

Challenge 1 — Classify the Source

EASY

Classify each example as internal / external and primary / secondary.

Example	Internal / External	Primary / Secondary
A survey your team designs for current customers		
Historical sales from your POS system		
A government unemployment dataset		
A purchased market report		
Interview notes from store managers		



Stretch Explain **one strength and one weakness** of each source for a customer-satisfaction project.

Challenge 2 — Choose a Research Design

MEDIUM

For each business question, decide whether observational or experimental data is more appropriate.

Business question	Preferred design	Why?
Does a new email subject line increase open rate?		
Are late deliveries associated with lower repeat purchase?		
Does staff training reduce complaint rate?		
Do rainy days affect store footfall?		



Stretch For one experimental question, identify the **treatment, control group and outcome variable**.

Challenge 3 — Convert the Management Problem

HARD

Management problem “Our premium product is underperforming.”

Required output	Your answer
Measurable analytical question	
Unit of analysis	
Target variable	
At least three data sources	
Measurement scale of at least three variables	
Recommended design: observational or experimental	
Success metric and business action	

ASSESSMENT · MULTIPLE CHOICE

End-of-Lecture Test — Part A

Question	A	B	C	D
1. Which DIKW level is raw facts with no context?	Data	Information	Knowledge	Wisdom
2. A government population dataset is usually:	Primary	Secondary	Experimental	Nominal only
3. Randomly showing two web pages to users is:	Observational	Experimental	Secondary	Qualitative
4. Product category is usually measured on which scale?	Nominal	Ordinal	Interval	Ratio
5. Revenue is usually measured on which scale?	Nominal	Ordinal	Interval	Ratio

Instructions Work individually for 5 minutes, then compare with a partner.

ASSESSMENT · SHORT ANSWER

End-of-Lecture Test — Part B

1. Define the DIKW model in your own words.
2. Give one internal and one external data source for analysing customer churn.
3. Explain the difference between primary and secondary data.
4. Why can observational data be useful but weak for causal claims?
5. Rewrite this vague problem as a measurable analytical question: **“Students are not engaging.”**

Submission format 3–5 bullet points per answer. Focus on **clarity and business examples**, not length.

ASSESSMENT

Answer Key and Marking Rubric

Part A answers

1. A — Data

2. B — Secondary

3. B — Experimental

4. A — Nominal

5. D — Ratio

Rubric for Part B — full credit requires

- **Q1:** a correct definition of all four levels, with a business example
- **Q2:** one clearly internal and one clearly external source
- **Q3:** the distinction based on **who collected it and for what purpose**
- **Q4:** a clear causal caution, ideally naming confounding
- **Q5:** a measurable question with **population, timeframe and metric**

SUMMARY

Key Takeaways

1. The **DIKW model** explains how raw facts become business decisions.
2. Data sources can be **internal or external**, **primary or secondary**.
3. Observational data is common; **experiments** give stronger causal evidence when feasible.
4. Variables must be **classified correctly** before analysis.
5. **Measurement scales** determine valid summaries, charts and models.
6. Strong projects start with a **specific, measurable, actionable question**.

Rule to remember **Do not start with data. Start with the business question.**

NEXT STEPS

Preparation for Chapter 3

Before the next lecture

- Read the short note on descriptive statistics.
- Bring one business question from your own experience.
- Install or confirm access to Python / Jupyter.
- Find one public dataset that interests you.

Next chapter: descriptive statistics

- Mean, median and mode
- Spread and variation
- Distributions and outliers
- Communicating summary statistics clearly

Closing prompt What **decision** could your chosen dataset help someone make?